



Docker Commands

Introduction

Docker provides **various commands** to manage containers, images, networks, and volumes.

- What are the essential Docker commands?
- How do these commands help in managing containers?
- How can you use Docker commands efficiently?

This guide covers **important Docker commands** with examples.

Section 1: Learn

Why Learn Docker Commands?

Feature	Benefit
Efficient Container Management	Start, stop, and remove containers easily
Image Management	Pull, build, and delete images
Networking	Connect and manage container networks
Volume Handling	Persist and manage data within containers

Categories of Docker Commands

Category	Purpose
Container Management	Create, start, stop, and remove containers



Category	Purpose
Image Management	Pull, build, tag, and remove images
Network Management	List, create, and inspect networks
Volume Management	Manage persistent storage for containers
System Commands	Check logs, processes, and events

Interesting Fact

Docker CLI commands follow a **simple and consistent syntax**, making it easy for developers to **automate deployments**.

Section 2: Practice

1. Container Management Commands

Run a New Container

```
docker run -d -p 8080:80 nginx
```

- **-d**: Runs the container in detached mode.
- **-p 8080:80**: Maps port 8080 on the host to port 80 in the container.

List Running Containers

```
docker ps
```



List All Containers (Including Stopped Ones)

```
docker ps -a
```

Stop a Running Container

```
docker stop <container_id>
```

Remove a Container

```
docker rm <container_id>
```

2. Image Management Commands

Pull an Image from Docker Hub

```
docker pull ubuntu
```

List Available Images

```
docker images
```

Remove an Image

```
docker rmi <image_id>
```

3. Network Management Commands

List Docker Networks

```
docker network ls
```



Inspect a Network

```
docker network inspect bridge
```

Create a New Network

```
docker network create my_network
```

4. Volume Management Commands

List Docker Volumes

```
docker volume ls
```

Create a New Volume

```
docker volume create my_volume
```

Remove a Volume

```
docker volume rm my_volume
```

5. System Commands

View Docker System Information

```
docker info
```

Check Running Processes in Docker

```
docker top <container_id>
```



Check Docker Logs for a Container

```
docker logs <container_id>
```

Restart Docker Daemon

```
sudo systemctl restart docker
```

Section 3: Know More

Frequently Asked Questions

1. What is the difference between **docker stop** and **docker kill**?
 - o **docker stop** gracefully shuts down the container, while **docker kill** immediately stops it.
2. How do I clean up unused Docker images and containers?

```
docker system prune
```

3. Can I rename a running container?
 - o Yes, use:

```
docker rename old_name new_name
```

4. How do I run an interactive container?

```
docker run -it ubuntu bash
```

5. How do I find the logs of a crashed container?

```
docker logs <container_id>
```



Conclusion

Docker commands **simplify container management** by allowing users to efficiently create, manage, and troubleshoot containers, images, networks, and volumes.

Start by **running containers, managing images, and automating deployments**, and soon you'll be **comfortable using Docker CLI commands for development and production!**